IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-154

1. Topic (12 words max)

Intelligent AI-Driven Pre-Construction Feasibility Analyzer for Integrated Construction Decision Support

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

In this segment, the project team should address the points such as:

- **Nature & Gravity of the Problem:**
  The pre-construction phase is the most critical stage of any infrastructure project, yet it remains highly fragmented and manual. The primary problem is the "Feasibility Gap." In Sri Lanka, inaccurate initial cost estimation is identified as a leading cause of project disputes and failure (Ramachandra & Rotimi, 2015). Currently, structural feasibility, cost estimation, and environmental impact are assessed in silos by different experts, leading to disconnected data and significant human error.
- **Technical & Economic Impact:**
  A major technical barrier is the low adoption of digital standards. Research by Jayasena & Weddikkara (2013) indicates that Building Information Modeling (BIM) adoption is minimal in the Sri Lankan construction industry due to high implementation costs. Consequently, the majority of the industry still relies on 2D blueprints (images/PDFs) and manual calculations. Existing global automation tools require 3D BIM models, rendering them useless for the local context.
- **Environmental Impact:**
  Furthermore, Environmental Impact Assessments (EIA) are typically conducted only after designs are finalized. Zubair (2001) highlights that this reactive approach often leads to regulatory delays or ecological damage that could have been avoided with early screening. There is no existing platform that integrates geometric, economic, and environmental feasibility into a single automated pre-design workflow.

- FoC Research Cluster and Industry Vertical/s : CoEAI – Centtre of Excellence for AI. Industry Vertical: Construction Technology (ConTech) & Real Estate (PropTech)

- How does the Problem attain Sustainable Development Goals (SDGs)?
- This research primarily targets **SDG 9 (Industry, Innovation, and Infrastructure)** by introducing AI-driven automation to modernize traditional, inefficient construction feasibility processes. Furthermore, it addresses **SDG 11 (Sustainable Cities and Communities)** and **SDG 13 (Climate Action)** by integrating a Predictive Environmental Impact Assessment (EIA) agent. This ensures that ecological risks (deforestation, high carbon footprint) are identified and mitigated at the very inception of a project, promoting sustainable urban planning.

- **References:**
  - **Ramachandra, T., & Rotimi, J. O. B. (2015).** Causes of projects cost overruns in the construction industry of Sri Lanka. International Journal of Construction Management.
  - **Jayasena, H. S., & Weddikkara, C. (2013).** Assessability of BIM in Construction Industry in Sri Lanka. The Second World Construction Symposium.
  - **Zubair, L. (2001).** Challenges for environmental impact assessment in Sri Lanka. Environmental Impact Assessment Review.

6. Brief description of Existing Research and Systems (300 words max)

In this segment, the project team should address the points such as:

- Summary of existing products/systems: Current industry standards rely on specialized, disconnected software.
  Cost Estimation: Tools like CostX or PlanSwift require manual tracing of digital drawings.
  Design & Modeling: Autodesk Revit and AutoCAD are powerful but require manual creation of 3D models.
  GIS: ArcGIS and Google Earth provide data but lack automated "constructability" scoring logic for specific building plots.

- Summary of existing research : Research exists in isolated domains. For instance, Sharma et al. developed CNNs for floor plan element detection, while others have used Remote Sensing for land cover classification. However, there is no existing framework that fuses **Computer Vision (for blueprints)** with **Satellite Analysis (for terrain)** to output a unified feasibility report

- Gaps/Limitations identified : **Lack of Integration:** No single platform assesses Geometric, Spatial, Economic, and Environmental feasibility simultaneously.
  BIM Dependency: Most modern automation tools require 3D BIM models. In Sri Lanka, 80%+ of projects still start with 2D drawings/images, rendering advanced tools useless.
  Skill Barrier: Conducting a full feasibility study requires a team of experts (QS, Engineer, Environmentalist), making it expensive and slow for small-to-medium projects.

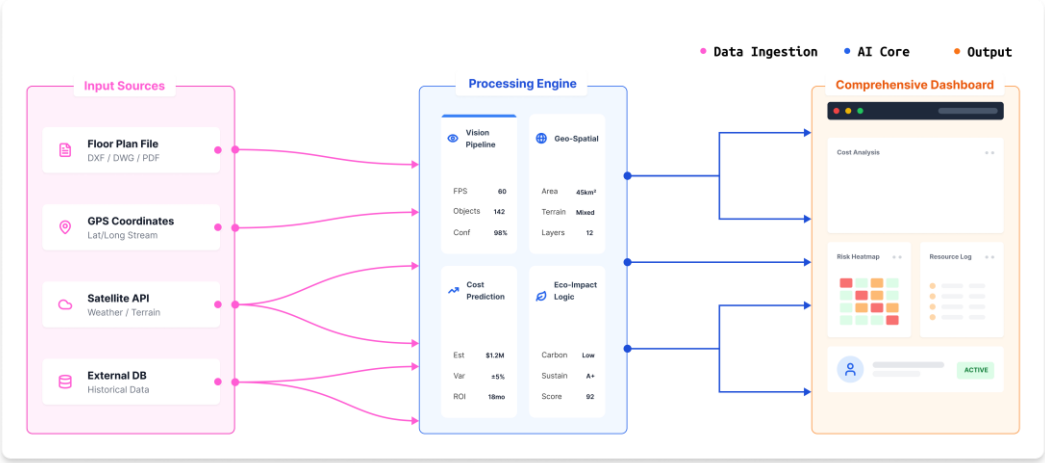
7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

In this segment, the project team should address the points such as:

- How does this system/research address gaps identified in Section 6 above?
  1. **Integration of Silos:** Unlike existing systems that treat cost, engineering, and environmental studies separately, this solution fuses **Visual Data** (Blueprints) and **Spatial Data** (Maps) into a single holistic assessment.
  2. **Overcoming BIM Unavailability:** While Western tools rely on 3D BIM models, this system is specifically designed for the Sri Lankan context. It utilizes **Computer Vision (CNNs)** to extract data from standard **2D image-based floor plans**, making automation accessible without expensive 3D modeling.
  3. **Proactive Environmental Assessment:** By employing a dedicated EIA Agent, the system shifts environmental analysis from a post-design regulatory step to a pre-design decision-making tool, reducing unforeseen ecological risks.

- System perspective of the Solution in the form of Input, Process, and Output.
  1. **Inputs:**
     - **Floor Plan Image:** User uploads a 2D image (JPG/PNG) of the proposed building layout.
     - **Geo-Location:** User pins the exact construction site on an interactive map interface.
  2. **Process (The 4-Agent Core):**
     - **Agent 01 (Geometric - U-Net):** Preprocesses the noisy floor plan image and performs semantic segmentation to extract "Quantities" (Floor Area, Wall Length, Room Count).
     - **Agent 02 (Spatial - ResNet/GIS):** Fetches satellite imagery for the location and classifies the terrain texture (e.g., Marshy, Rocky, Flat) to determine "Site Difficulty."
     - **Agent 03 (Economic - XGBoost):** A predictive regression model that combines the Quantities (from Agent 1) and Site Difficulty (from Agent 2) with CIDA market rates to forecast Total Cost and Duration.
     - **Agent 04 (Environmental - Fuzzy Logic):** Uses rule-based logic to calculate the Carbon Footprint and Environmental Risk Score based on material volume and land sensitivity.
  3. **Output:**
     comprehensive **Feasibility Dashboard** displaying:
     - Estimated Budget (LKR) & Project Duration.
     - Site Suitability Index (0-100%).
     - Environmental Impact Rating (Low/Medium/High).

- *Include a clear diagram with legible fonts (if space is not enough, use a separate page).*



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

In this segment, the project team should address the points such as:

- How can members gain essential specialized domain expertise?
  Acquisition of Specialized Domain Expertise - Since this project addresses problems in the Construction and Civil Engineering domain, the research group will adopt a hybrid approach to gain the necessary domain knowledge:

  - **Expert Consultation:** The team will maintain regular contact with external industry experts, specifically **Chartered Quantity Surveyors (IQSSL)** and Civil Engineers. This is crucial to understand the manual logic behind "Taking-off" quantities and determining site constructability.
  - **Standard Documentation:** We will utilize the **CIDA (Construction Industry Development Authority)** publications, specifically the Standard Method of Measurement (SMM7) and Building Schedule of Rates (BSR), which govern construction estimation in Sri Lanka.
  - **Environmental Guidelines:** For the environmental component, the team will refer to the **Central Environmental Authority (CEA)** guidelines on EIA procedures to derive accurate Fuzzy Logic rules.

- How can a member gain the technical knowledge and skills required for the project?
    1. **Online Coursework:** Members will complete specialized courses on platforms like Coursera and Udemy.
    2. **Literature Review:** We will study recent IEEE and Springer research papers on "Automated Floor Plan Analysis" and "Remote Sensing for Construction" to identify state-of-the-art algorithms.

- How is the data required for the project collected?
  We will collect the data required for this project from three main sources:
  Floor Plan Images: To train the AI to read blueprints, we will use a public dataset called **CubiCasa5k**. To make sure the system understands Sri Lankan house styles, we will also collect floor plan images from local websites.

  Satellite & Map Data: To analyze the land terrain remotely, we will get satellite images using the **Google Maps API** and **Sentinel-2** satellite data. We will also use **OpenStreetMap** to get information about nearby roads and water bodies.

  Cost & Environmental Data: For accurate cost estimation, we will use the official **CIDA Building Schedule of Rates (BSR)**, which provides the government-approved prices for materials and labor in Sri Lanka. For environmental analysis, we will use standard databases that list the carbon footprint of building materials.

- Does this project require ethical clearance?
  The research deals strictly with **inanimate objects** (building blueprints), **geographical data** (land plots), and **public cost data**. It does not involve human participants, biological data, or sensitive personal information. All used images are from open datasets or public listings where privacy is already managed.

9. Objectives and Novelty

Main Objective To develop an integrated Multi-Agent AI platform that automates the pre-construction feasibility process by synthesizing 2D floor plan images and satellite terrain data to provide instant assessments on geometric quantities, site suitability, construction cost, and environmental impact.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Shamindi E E IT22547460	To develop a smart prediction engine that calculates the project cost based on location difficulty and suggests material changes to save money.	Data Integration: Combining the measurements from Member 01 and the terrain type from Member 02 into one dataset. Smart Cost Prediction: Using a Machine Learning model to predict the budget. It increases the price automatically if the land is Material Optimization: Creating a logic system that suggests cheaper or better alternatives if the estimated cost is too high.	Unlike standard calculators that just multiply Area by Rate, this is a **"Prescriptive Model."** It is novel because it acts like a consultant: it adjusts the price based on the specific location's difficulty and proactively suggests **ways to reduce the cost**, which normal software cannot do.

Ranaweera K A IT22542724	To build a remote sensing AI system that uses satellite images to identify the type of land (terrain) and determine if it is suitable for construction without a physical visit.	Satellite Data Collection: Fetching high-quality images of the specific land location using Google Maps and Sentinel satellite APIs. Terrain Classification: Using a Deep Learning model to look at the ground texture and classify it as Rocky, Marshy, Flat, or Vegetation. Constructability Scoring: Generating a "Difficulty Score" for the land. For example, giving a low score for marshy land because it is hard to build on.	The novelty is the concept of a **"Virtual Site Visit."** Usually, engineers must visit the land to see if it is marshy or rocky. This system uses texture analysis of satellite maps to identify these risks remotely, saving time and transport costs.
Shavinka Y G K IT22110770	To develop a Computer Vision model that can read 2D floor plan images (even low-quality ones) and automatically calculate the floor area and wall lengths needed for quantity surveying.	Image Pre-processing: Applying filters to clean up scanned or noisy images so the AI can read them clearly. Semantic Segmentation: Training a U-Net AI model to identify pixels that represent walls, rooms, and doors in the image. Quantity Extraction: Converting the identified pixels into real-world measurements (like square feet and linear meters) to be used for cost calculation.	Most existing tools require perfect digital files (CAD or BIM). This solution is novel because it works on **noisy, scanned, or hand-drawn images**, which are very common in Sri Lanka. It bridges the gap between paper drawings and digital estimation.

Samuditha T A V IT22542892	To predict the environmental impact (like Carbon Footprint) of a building at the very beginning stage using Fuzzy Logic rules.	Rule Creation: Setting up "If-Then" rules (Fuzzy Logic) to decide how much damage a building causes based on its size and location. Carbon Calculation: Estimating the total Carbon Footprint by analyzing the amount of concrete and other materials needed. Risk Reporting: Generating an automatic report that warns the user if the project has a "High Environmental Risk" or causes too much deforestation.	Currently, Environmental Impact Assessments (EIA) happen at the end of the design process. The novelty here is doing a **"Pre-design Environmental Check."** It warns the user about ecological damage before they even start the detailed design, encouraging sustainable building.

10. <u>This part is to be filled by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

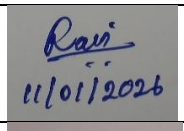
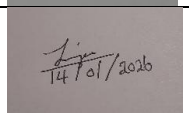
c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Mr	Ravi	Supunya	
Co-Supervisor	Ms	Suriya	Kumari	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				